23. Applied Engineering Mathematics III

<table>
<tr><th colspan="5">Program of Electrical & Computer Engineering
University of Arba Minch , Arba Minch Institute of Technology</th></tr>
<tr><td colspan="5"></td></tr>
<tr><td>Course Number</td><td colspan="4">Math2121</td></tr>
<tr><td>Course Title</td><td colspan="4">Applied Engineering Mathematics III</td></tr>
<tr><td>Degree Program</td><td colspan="4">B. Sc. in Electrical Engineering</td></tr>
<tr><td>Module</td><td colspan="4">Advanced Engineering Mathematics</td></tr>
<tr><td>CP Credits</td><td colspan="4">6 CP</td></tr>
<tr><td rowspan="2">Contact Hours (per week)</td><td>3</td><td>Practice or Laboratory</td><td>Tutorial</td><td>Home study</td></tr>
<tr><td>2</td><td>0</td><td>3</td><td>5</td></tr>
<tr><td>Course Objectives</td><td colspan="4">Students shall understand the principles of differential equations of 1st and 2nd order, vector differential calculus, line & surface integral, complex analytical functions, complex integrals, different types of series</td></tr>
<tr><td>Course Description/Course Contents</td><td colspan="4">• First order differential equations
• Second order differential equations
• Vector differential calculus
• Line and surface integral
• Complex analytical functions and complex integrals
• Taylor and Laurant Series
• Integration by the method of residue
• Complex Number and Electrical Applications
• Four Analysis: Fourier Series and Four Transform
• Z Transform and System Analysis</td></tr>
<tr><td>Pre-requisites</td><td colspan="4">Applied mathematics II</td></tr>
<tr><td>Semester</td><td colspan="4">Year 2, Semester II</td></tr>
<tr><td>Status of Course</td><td colspan="4">Compulsory</td></tr>
<tr><td>Teaching & Learning Methods</td><td colspan="4">Lectures, tutorials</td></tr>
</table>

Assessment/Evaluation & Grading System	Refer Article 9 of the "*Academic Regulations for Engineering Programs*"
Attendance Requirements	A student must attend at least 85 % of lecture class and 100% of tutorial class.
Literature	**1.** Erwin Kreyszig (2005). Advanced Engineering Mathematics, 9th edition, Wiley **2.** Ellis, R. and Gulick, D. (1998). Calculus with Analytic Geometry, 5th edition. Harcourt **3.** Stewart, J. (2002), Calculus, 5th edition, Brooks Cole **4.** Churchil, R. V. (2003). Complex Variables and Application, 7/e, McGraw Hill Edition

24. Computational Methods

Program of Electrical & Computer Engineering **University of Arba Minch , Arba Minch Institute of Technology**				
Course Number	ECEg3122			
Course Title	Computational Methods			
Degree Program	B. Sc. in Electrical Engineering			
Module	**Mathematics and Physics**			
CP Credits	**6 CP**			
Contact Hours (per week)	Lecture	Practice or Laboratory	Tutorial	Home study
	3	3	0	5
Course Objectives	Students shall learn basics of mathematical modeling, different numerical methods for determination of roots of equations, fundamentals of linear algebraic equations, least square regressions and interpolation methods, numerical differentiation and integration, and solving ordinary differential equations numerically.			